

Dear Professor Marc & Selection Team,

I am writing to apply for the EngD position on the **interactive reverse logistics map** for circular construction (CIRCOLOGIC). In construction we are already good at taking buildings apart, but a lot of good material still ends up as waste. From what I understand, the main problem is information. People often do not know which materials are available, in what condition, where they are, or when they will need them. So they choose new materials instead of reused ones. A map that shows what is available, a way to calculate cost and CO₂, and a simple way to compare options would make reuse a much easier choice. That is the kind of solution I would like to help build.

My education gives me a useful base for this. I hold a Master's in Environmental Engineering and a Bachelor's and Diploma in Civil Engineering. In my master's thesis, **Agile Project Management in the Construction Industry**, I studied how Agile can help construction teams work better together, manage risk, make decisions and use resources more carefully. When I finished, I felt something was still missing. Better delivery and less waste on site only cover part of sustainability. I started asking myself what happens to materials when a building reaches the end of its life. That led me to circular construction and reverse logistics, and to earlier work such as Professor Voordijk's research on recovering building elements for reuse. One point stayed with me: reuse needs coordination from the donor building to the target building, not only new technology. An interactive reverse logistics map is a step further than logistics and Circular Matching Platforms like DuSpot, especially when timing, quality and cost are uncertain.

My work experience adds practical skills I can bring to the project. At Aug.e I worked on digital twins for smart buildings. We used a digital model to understand and improve how a real building performs. At Hive CPQ I managed large product datasets from manufacturers and turned complex information into simple tools that non-technical users could work with. At iostriing I worked with a construction partner on a circular construction hypothesis: how Agile project management can bring circular practices into real projects. We did not reach a full pilot. Across these roles I learned to work with construction data, stakeholders and digital tools. Together with my GIS skills, this is the expertise I can offer from day one. I have been on a career break to care for my newborns, and I kept learning about this field during that time.

At the same time, I am motivated by what I still want to learn. I want to grow in reverse-logistics modelling, use GIS more carefully to match donor and target materials, and learn how to build and test a decision-support map with industry partners. The EngD at the University of Twente is a strong fit for that, because it mixes advanced study with a real design project. I would like to work in a design-science way: start from real use cases, test the map in a (de)construction pilot with partners like Dura Vermeer and TNO, and improve it with the people who will use it. I do not want to build another tool that sits unused. I want to help build something people will actually use. I can relocate to the Netherlands.

Thank you for reading my application. I would be glad to talk further.

Yours sincerely,

Pooja Awasare